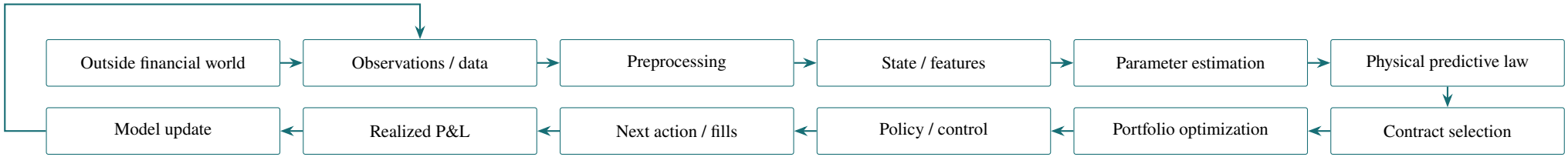
Multi-asset futures: data, thesis and path law

One-period baseline retained; optional slow thesis state and multi-period control use the same gain and cash maps.



Observations update beliefs; only decision dates (or mandatory risk events) trigger strategic orders. Execution has its own faster clock.

Three clocks. Observations arrive at o_j; decisions occur at t_k with spacing Δ_d; the economic thesis expires at a fixed T^{th} (e.g. weeks). Planning depth H counts decision intervals, $t_{k+H} \le T^{\text{th}}$; it is not the observation count. Below k indexes decisions and $\mathcal{F}_k$ contains only released information. Freeze training length N, PCA rank $r < n$, scales, priors, calendar/surprise laws, risk/cost bounds, seed and validation splits. All money uses one currency. Do not roll the thesis deadline forward after losses.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F0. Data, gains, coordinates $O_{\le k}$: dated records; $g_s, x_s \in \mathbb{R}^n$. $D \in \mathbb{R}^{n\times n}$ positive diagonal; $\bar r \in \mathbb{R}^n$. $B = [V\ U] \in \mathbb{R}^{n\times n}$ orthogonal, $V \in \mathbb{R}^{n\times r}$.	Released settlements, multipliers m_i, liquidity, funding, margin and fills enter $g_{i,s} = m_i(F_{c_i(s-1),s} - F_{c_i(s-1),s-1})$, using the maturity held over that interval. Never splice a roll price jump into gain. On N training gains, $D_{ii} = \mathrm{sd}(g_i) > 0$, $r_s = D^{-1}g_s$, $\bar r = N^{-1}\sum r_s$. Eigenvectors of $S_r = (N-1)^{-1}\sum(r_s - \bar r)(r_s - \bar r)^\top$ give B, ordered with fixed signs/ties. Fast factor/residual coordinates: $x_s = B^\top(r_s - \bar r)$. Freeze/recompute the whole block in one basis. **To F1–F3, F5.**	NumPy data `(N,n)`, basis/covariance `(n,n)`, state `(n,)`; timestamped contract records.
F1. One-step baseline fit $\widehat A \in \mathbb{R}^{n\times n}$, $\widehat Q \in \mathbb{S}^n_{++}$; $\mu_k \in \mathbb{R}^n$, $\Sigma_k \in \mathbb{S}^n_{++}$. $\widehat\theta$: all fitted coefficients.	$x_{s+1} = Ax_s + \varepsilon_{s+1}$, $\varepsilon \sim \mathcal{N}(0, Q)$. $\widehat A = (\sum x_{s+1}x_s^\top)(\sum x_s x_s^\top + \lambda I)^{-1}$, $\widehat Q = (N-1)^{-1}\sum(x_{s+1} - \widehat A x_s)(x_{s+1} - \widehat A x_s)^\top + \epsilon I$, $\lambda, \epsilon > 0$. Reject spectral radius $\rho(\widehat A) \ge r_* < 1$. Baseline $g_{k+1} \mid \mathcal{F}_k \sim \mathcal{N}(\mu_k, \Sigma_k)$, $\mu_k = D(\bar r + B\widehat A x_k)$, $\Sigma_k = DB\widehat Q B^\top D$. **To F3, F5; this is one step, not a thesis-horizon forecast.**	Arrays `(n,n)`, `(n,)`; fit/transition callables; SciPy Gaussian law.
F2. Slow thesis extension $u_k \in \mathbb{R}$: latent valuation anchor; $\ell_k \in \mathbb{R}$: observed reference-spread level. $d_k = u_k - \ell_k$: remaining gap; $y_k \in \mathbb{R}$: economic proxy. $c_{k+1} \in \mathbb{R}^p$: catalyst surprises; $b_k = \mathrm{Law}(u_k \mid \mathcal{F}_k)$.	Choose a fixed reference basket $w_0 \in \mathbb{R}^n$, capital $A_0 > 0$, and a predeclared economic valuation proxy y in the same units: $\ell_{k+1} = \ell_k + w_0^\top g_{k+1}/A_0$, $u_{k+1} = \bar u + \phi(u_k - \bar u) + \beta^\top c_{k+1} + \sigma_u\eta_{k+1}$, $y_k = u_k + \tau_y\nu_k$, $x_{k+1} = Ax_k + a_d d_k + J_c c_{k+1} + \varepsilon_{k+1}$. Here $a_d \in \mathbb{R}^n, J_c \in \mathbb{R}^{n\times p}, \beta \in \mathbb{R}^p$, $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$; independent standardized Gaussian η, ν. Persistence is estimated ϕ near one and small σ_u, not immunity to evidence. At announced releases $c \sim \mathcal{N}(\bar c, C_c)$, $C_c \in \mathbb{S}^p_+$, fitted to past surprises; otherwise $c = 0$. Filter from a proper prior: $b_{k+1}(u) \propto p(y_{k+1} \mid u)\int p(x_{k+1}, u \mid x_k, u', \ell_k) b_k(u')\,\mathrm{d}u'$; maximize predictive log likelihood (EM) for coefficients. Between decision-grid VAR observations, released y's update the current anchor likelihood; quotes update risk. $a_d = J_c = 0$ recovers F1. **To F3–F6.**	Filter mean/variance `(1,)`, `(1,1)`; coefficients `(n,p)`, `(p,)`. Proxy map and catalyst calendar fixed before testing.
F3. Joint physical path law $Z_k = (x_k, u_k, \ell_k) \in \mathbb{R}^{n+2}$; K^P_θ: F1/F2 transition. $\mathsf{P}^H_k \in \mathcal{P}((\mathbb{R}^{n+2})^{H+1})$. $L_k \in \mathbb{R}^{n\times n}$, $\rho_\mu \ge 0$: mean uncertainty.	$\mathsf{P}^H_k(\mathrm{d}Z_{k:k+H}) = b_k(\mathrm{d}u_k)\prod_{j=k}^{k+H-1} K^P_{\widehat\theta}(\mathrm{d}Z_{j+1} \mid Z_j)$, observed x_k, ℓ_k fixed, catalysts integrated. The endpoint $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only a marginal. For F1 alone it gives $x_{k+H} \sim \mathcal{N}(A^H x_k, \sum_{j=0}^{H-1} A^j Q (A^j)^\top)$; cumulative gain and interim risk need cross-time dependence. Use $\mu_k = \mathbb{E}^P_k g_{k+1}$, $\Sigma_k = \mathrm{Cov}^P_k(g_{k+1})$, including anchor uncertainty. Block-bootstrap refits give $\mu_k^{(b)}$, $L_k L_k^\top = \widehat{\mathrm{Cov}}_b(\mu_k^{(b)}) + \epsilon_\mu I$; ρ_μ is a frozen quantile of $\|L_k^{-1}(\mu_k^{(b)} - \mu_k)\|$. Hold each fitted parameter set fixed per path. **To F5–F6.**	Paths `(N_paths,H+1,n+2)`, gains `(N_paths,H,n)`; Gaussian distribution or simulation callable; bootstrap means `(N_boot,n)`.
F4. Thesis status / expression $p_k = b_k(u_k - \ell_k > \delta) \in [0,1]$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed. $p_{\text{on}} > p_{\text{off}}, \delta, \delta_{\text{exit}} \ge 0$; eligible mask $j_k \in \{0,1\}^n$.	For a positive-gap thesis, activate at $p_k \ge p_{\text{on}}$; retain it between thresholds. Close on convergence $\mathbb{E}_k d_k \le \delta_{\text{exit}}$, $p_k \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or T^{th}; freeze thresholds and label reasons. Waiting forbids new thesis risk; † cancels entry orders and flattens, and is absorbing until a new thesis record. Risk/margin failures act immediately. The thesis is u, d, b, not a contract ID: choose maturities, spreads, hedges or zero anew. Shares/options require their own pricing/ledger. Eligibility $j_i = 1$ requires valid liquid quotes, capped spreads and dates before notice/roll cutoff; held ineligible legs exit. Disabling this extension uses F1/F5 without the thesis gate. **To F5–F6.**	Scalar probabilities, mode/deadline record, Boolean mask `(n,)`. Log thesis exit separately from temporary zero exposure.

Futures: strategic control, execution and accounting

The baseline uses H=1. Longer planning adds continuation value; observing another tick does not itself authorize another position change.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
F5. Baseline target / constraints $s^{\text{acct}} = (q^-, X, M) \in \mathbb{Z}^n \times \mathbb{R}^2$, $W = X + M$. Target $q \in \mathbb{Z}^n$, roll map $\mathcal{R}(q) \in \mathbb{Z}^{2n}$. $s, \eta \in \mathbb{R}^{2n}_+$, $\ell^M \in \mathbb{R}^n_+$: spread/impact/margin.	With diagonal roll flags R^{roll}, $a = \mathcal{R}(q) = (-R^{\text{roll}}q^-, q - (I - R^{\text{roll}})q^-)$. Read spreads/margin; fit nonnegative impact from fills. Define $K(a) = s^\top\lvert a\rvert + \sum \eta_i a_i^2$, $M^+(q) = (\ell^M)^\top\lvert q\rvert$, $d^c(q) = (R^M - R^f)M^+(q) - R^f K(a)$, $v(q) = q^\top \Sigma_k q$. Baseline score $J_1(q) = [q^\top\mu_k - \rho_\mu\|L_k^\top q\| + d^c(q)]/A_0 - \gamma v(q)/(2A_0^2)$. Feasible C_k: $\lvert q_i\rvert \le j_i \bar{q}_i$, $\|Dq\|_1 \le L_g$, $\lvert V^\top Dq\rvert \le b$, $\lvert H_g Dq\rvert \le c_g$, $W - K - M^+ \ge R_{\text{cash}}$, stress-equity floors, and $-q^\top\mu + \rho_\mu\|L^\top q\| - d^c + \sqrt{v}\,\varphi(\Phi^{-1}(\alpha))/(1-\alpha) \le L_{\text{ES}}$. This Gaussian ES formula applies only to the one-step Gaussian law; use simulated loss ES for longer paths. **To F6.**	CVXPY integer vector `(n,)` or bounded enumeration; constraints use supplied caps, $H_g \in \mathbb{R}^{p_g \times n}$. Gross factors $R^f, R^M > 0$; normal PDF/CDF φ, Φ.
F6. Bellman / stochastic MPC $s_k = (\mathcal{F}_k$ summary, $b_k, z_k, T^{\text{th}} -$ t_k, s_k^{acct}, pending orders). $V^h, Q^h : \mathcal{S} \to \mathbb{R}$, $\mathcal{S} \times \mathcal{A} \to \mathbb{R}$. $\eta_T, h_0 \ge 0$: turnover/switch preferences.	Let $\Pi = W' - R^f W$ from F7 and $r_F = \Pi/A_0 - \gamma\,\mathrm{Var}_k(\Pi)/(2A_0^2) - \rho_\mu\|L_k^\top q\|/A_0 - \eta_T\|a\|_1$. $Q_k^h(s,a) = \mathbb{E}_k^P[r_F + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}_k(s)}\{Q_k^h(s,a) - h_0 \mathbf{1}_{a \ne 0}\}$. Choose $V^0 = 0$ to recover F5 at $H = 1, \eta_T = h_0 = 0$; use a specified continuation value at a shorter planning boundary, and $-K^{\text{liq}}/A_0$ at the thesis endpoint. Simulate proxy observations and re-filter at future nodes: actions never see latent states or later shocks. Stochastic MPC executes the first action of this nonanticipative tree; full Bellman control uses the remaining horizon. No-action region: $\max_{a \ne 0}[Q^H(s,a) - Q^H(s,0)] \le h_0$, if holding is feasible. F4 hysteresis and this band suppress reversals; mandatory exits/rolls override them. **To F7.**	Scenario tree / value callable; finite belief representation $\varphi_s(s) \in \mathbb{R}^{d_s}$. Costs are already in Π; turnover and switch penalties are preferences, not second cash debits.
F7. Separate execution / ledger $a^{\text{str}} \in \mathbb{Z}^{2n}$: strategic parent order; $y_e, v_e \in \mathbb{R}^{2n}$: remaining order/rate. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{2n \times 2n}$, $T_e \ll \Delta_d$, $\lambda_e \ge 0$.	Execution chooses $v_e = -\dot{y}_e$ to minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $y_e(0) = a^{\text{str}}$, $y_e(T_e) = 0$, subject to lot/rate/interim-risk limits. Fit impact Λ_e from fills and quote-change covariance Ω_e; carry unfilled quantities if completion fails. Feed expected cash cost back into F5/F6, replacing its cost estimate; execution-risk penalty is separate. Full immediate fills remain the baseline: $q^+ = q^- + \tilde{a}^{\text{old}} + \tilde{a}^{\text{new}}$, $X^+ = X + M - M^+(q^+) - K(\tilde{a})$, $W' = R^f X^+ + R^M M^+(q^+) + (q^+)^\top g'$, $\Pi = W' - R^f W$. Reset $M' = (\ell^{M'})^\top\lvert q^+\rvert$, $X' = W' - M'$. With gradual fills, accrue cash event by event and use $\int q(u)^\top\,\mathrm{diag}(m)\,\mathrm{d}F(u)$ over actual maturities. **To next account/filter.**	Execution optimizer/callable; order/fill arrays `(2*n,)`; event ledger. Strategic controller chooses exposure; execution chooses timing/size of child orders.
F8. Optional RL; when / interface $Q_\psi : \mathbb{R}^{d_s+2n} \to \mathbb{R}$: learned action value; $\bar{\psi}$: target parameters.	Input the full finite belief/account/mode/deadline encoding and parent action. Train from supported transitions (s, a, r_F, s') on $\mathcal{L}(\psi) = \mathbb{E}[(Q_\psi(s,a) - \{r_F - h_0\mathbf{1}_{a \ne 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')\})^2]$, replacing the maximum by terminal value at the endpoint. Use RL when repeated large control solves need approximation, or action-dependent fills/information must be learned from adequate data. It is unnecessary when the fitted law and constrained MPC/Bellman solve are tractable. Validate simulator/offline coverage and preserve hard constraints; RL cannot supply an unidentified thesis edge. **Replaces F6 value solve only.**	Model callable / tensors; sufficient belief encoding (not an arbitrary posterior mean), feasible-action mask, replay transitions. Unseen actions require evidence, not extrapolation claims.
F9. Validation / gates $d_j^{\text{test}} = (\Pi_j - \Pi_j^{\text{bench}})/A_0$; J test decisions; $\widehat{\Gamma}_l$: sample lag covariance.	Chronological train/select/test blocks; test joint-path calibration, thesis persistence, catalyst forecasts, turnover, execution and net gains against frozen benchmarks. $Z = \sqrt{J}(\bar{d}^{\text{test}} - \delta_{\text{econ}})/\sqrt{\widehat{\Omega}}$, $\widehat{\Omega} = \widehat{\Gamma}_0 + 2\sum_{l=1}^{L_v}(1 - l/(L_v+1))\widehat{\Gamma}_l$. Predeclare thresholds/multiplicity; resample blocks preserving overlap and longer observed dependence. Gate solves only for valid inputs and nonempty feasible sets; otherwise funded liquidation or default/stop. Save posterior, gap, value difference and gate reason. **To next F0 and action gate.**	Scores `(J,)`, path diagnostics, block resampling; mode/risk/model/execution exits have distinct reason codes.

Composition: $O_{\le k} \to (x_k, \ell_k, y_k) \to (\widehat{\theta}, b_k, d_k, z_k) \to \mathsf{P}_k^H \to (Q_k^H, \mathcal{A}_k) \to a_k^{\text{str}} \to$ child fills $\to W_{k+1} \to b_{k+1}$.

$$a_k^* = F(O_{\le k}, s_k^{\text{acct}}, \widehat{\theta}) = \mathcal{E}_{\text{F7}}\left[\text{Gate}_{\text{F4,F9}}\left(\arg\max_{a \in \mathcal{A}_k(s_k)}\left\{\mathbb{E}_{\mathsf{P}_k^H}[r_F(s_k, a, s') + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a \ne 0}\right\}\right)\right].$$